[illegible]

"[illegible]" [illegible]

[illegible]

[illegible]

[illegible], [illegible]:

1. [illegible]: [illegible] A, [illegible] $A \subseteq A$.
2. [illegible]: [illegible] A, [illegible] $\emptyset \subseteq A$.
3. [illegible]: [illegible] A, B:
   - $A \subseteq A \cup B$
   - $A \cap B \subseteq A$

?[illegible]"[illegible] [illegible] [illegible]

- :

$$A \cup B = \{x \mid x \in A \text{ [illegible] } x \in B\}$$

- :

$$A \cap B = \{x \mid x \in A \text{ [illegible] } x \in B\}$$

- [illegible] [illegible]:

$$A \setminus B = \{x \mid x \in A \text{ [illegible] } x \notin B\}$$

- [illegible] [illegible]:

$$A \Delta B = (A \cup B) \setminus (A \cap B)$$

- [illegible] [illegible]:

$$A \times B = \{\langle x, y \rangle \mid x \in A \text{ [illegible] } y \in B\}$$

- [illegible] [illegible]:

$$\mathcal{P}(A) = \{X \mid X \subseteq A\}$$

- [illegible] [illegible]/[illegible]:

$$\bigcup \mathcal{F} = \{x \mid \exists X \in \mathcal{F}(x \in X)\}$$

$$\bigcap \mathcal{F} = \{x \mid \forall X \in \mathcal{F}(x \in X)\}$$

- :

$$R \subseteq A \times B$$

- [illegible] [illegible]:

$$\text{dom}(R) = \{x \mid \exists y(\langle x, y \rangle \in R)\}$$

$$\text{img}(R) = \{y \mid \exists x(\langle x, y \rangle \in R)\}$$

[illegible] :[illegible]

	([illegible]) [illegible]	([illegible]) [illegible]
$\alpha \to \beta$	β [illegible] ,α [illegible]	(MP) β [illegible] ,α [illegible]
$\alpha \vee \beta$	β [illegible] $\neg\alpha$ [illegible]	β 2 [illegible] ,α 1 [illegible] :[illegible]
$\forall x P(x)$	$P(x)$ [illegible] .[illegible] x [illegible]	[illegible] c [illegible] $P(c)$ [illegible]
$\exists x P(x)$	$P(c)$ [illegible] c-[illegible] [illegible]	(![illegible]) $P(c)$ [illegible] c [illegible]
$A = B$	$B \subseteq A$ [illegible] $A \subseteq B$ [illegible]	([illegible]) B-[illegible] A [illegible]

[illegible]

[illegible]: [illegible] $\mathcal{F}$ [illegible] A:

$$A \setminus \bigcup \mathcal{F} = \bigcap \{A \setminus X \mid X \in \mathcal{F}\}$$

$$A \setminus \bigcap \mathcal{F} = \bigcup \{A \setminus X \mid X \in \mathcal{F}\}$$

[illegible]:

$$\begin{aligned}
A \setminus \bigcup \mathcal{F} &= \{x \mid x \in A \land x \notin \bigcup \mathcal{F}\} \\
&= \{x \mid x \in A \land \neg(x \in \bigcup \mathcal{F})\} \\
&= \{x \mid x \in A \land \neg(\exists X \in \mathcal{F}(x \in X))\} \\
&= \{x \mid x \in A \land \forall X \in \mathcal{F}(x \notin X)\} \\
&= \{x \mid \forall X \in \mathcal{F}(x \in A \setminus X)\} \\
&= \bigcap \{A \setminus X \mid X \in \mathcal{F}\}
\end{aligned}$$

[illegible]: [illegible]

[illegible]: [illegible]:

$$((p \to q) \land (q \to r)) \to (p \to r)$$

[illegible] ([illegible]). [illegible]. [illegible]:

1. [illegible]. [illegible].
2. [illegible] $(\neg)$ $\neg(p \to r) \Longleftarrow p = T, r = F \Longleftarrow$.
3. [illegible] $p \to q \Longleftarrow$ [illegible] $q \to r$ [illegible].
4. [illegible]-$p = T$ [illegible]-$p \to q$ [illegible] $q = T \Longleftarrow$.
5. [illegible]-$q = T$ [illegible]-$q \to r$ [illegible] $r = T \Longleftarrow$.
6. [illegible] $r = T$ [illegible] $r = F$ ([illegible]).

[illegible]: [illegible]

[illegible]: [illegible] "[illegible]" [illegible]:

$$\forall x \exists y (P(x) \to Q(x,y))$$

[illegible]: [illegible]:

1. $\neg \forall x \exists y (P(x) \to Q(x,y))$
2. $\exists x \neg \exists y (P(x) \to Q(x,y))$ ([illegible])
3. $\exists x \forall y \neg (P(x) \to Q(x,y))$ ([illegible])
4. $\exists x \forall y (P(x) \land \neg Q(x,y))$ ([illegible])

[illegible]: $\exists x \forall y (P(x) \land \neg Q(x,y))$

A, B:

$$A \Delta B = (A \cup B) \setminus (A \cap B)$$

$$\begin{aligned}(A \cup B) \setminus (A \cap B) &= \{x \mid x \in (A \cup B) \land x \notin (A \cap B)\} \\ &= \{x \mid (x \in A \lor x \in B) \land \neg(x \in A \land x \in B)\} \\ &= \{x \mid (x \in A \lor x \in B) \land (x \notin A \lor x \notin B)\} \\ &= \{x \mid (x \in A \land x \notin B) \lor (x \in B \land x \notin A)\} \\ &= \{x \mid x \in (A \setminus B) \lor x \in (B \setminus A)\} \\ &= (A \setminus B) \cup (B \setminus A)\end{aligned}$$

[illegible]

[illegible]: [illegible] $A = \{1, 2\}$.

1. [illegible] $\mathcal{P}(A)$ ([illegible] A).
2. [illegible] $\{1\} \in \mathcal{P}(A)$?
3. [illegible] $\{1\} \subseteq \mathcal{P}(A)$?

[illegible]:

1. $\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.
2. [illegible]. $\{1\}$ [illegible] A, [illegible] $\mathcal{P}(A)$-[illegible].
3. [illegible]. [illegible] $\{1\} \subseteq \mathcal{P}(A)$-[illegible], [illegible] 1 [illegible] $\mathcal{P}(A)$-[illegible], [illegible] 1 [illegible] ([illegible]), [illegible] $\mathcal{P}(A)$ [illegible].

□□□□: □□□□□□□ □□ □□□□□□□:

$$A \times (B \setminus C) = (A \times B) \setminus (A \times C)$$

□□□□□□:□□□□. (□□□□□ □□□□□□ 02, □□□ □□□ □□□□□□□ □□□□□□□): □□□ $\langle x, y \rangle$.

$$\begin{aligned}
\langle x, y \rangle \in A \times (B \setminus C) &\iff x \in A \wedge y \in (B \setminus C) \\
&\iff x \in A \wedge (y \in B \wedge y \notin C) \\
&\iff (x \in A \wedge y \in B) \wedge (x \in A \wedge y \notin C) \\
&\iff \langle x, y \rangle \in A \times B \wedge \neg(x \in A \wedge y \in C) \\
&\iff \langle x, y \rangle \in A \times B \wedge \langle x, y \rangle \notin A \times C \\
&\iff \langle x, y \rangle \in (A \times B) \setminus (A \times C)
\end{aligned}$$

(□□□□□□□ □□□□□ □□□□□□□ □□□□□□□: □□ $x \in A$, □□ $y \in C \iff \langle x, y \rangle \in A \times C$. □□ $x \notin A$, □□□ □□□□□□□ □□□□□□□.)

[illegible]

[illegible] 6: [illegible] $R = \{\langle 1,2\rangle, \langle 2,3\rangle, \langle 3,1\rangle\}$ [illegible] $A = \{1,2,3\}$.

1. [illegible] $dom(R)$?
2. [illegible] R^{-1}?
3. [illegible] R [illegible]? [illegible] R [illegible]-[illegible]?

[illegible]:

1. [illegible]: $dom(R) = \{1,2,3\}$ ([illegible]).
2. [illegible]: [illegible].

$$R^{-1} = \{\langle 2,1\rangle, \langle 3,2\rangle, \langle 1,3\rangle\}$$

3. [illegible]:
   - [illegible] - $\langle 1,2\rangle, \langle 2,3\rangle \in R$ [illegible] $\langle 1,3\rangle \notin R$.
   - [illegible]-[illegible] [illegible] - [illegible] $x \neq y$ [illegible] $\langle x,y\rangle \in R$ [illegible] $\langle y,x\rangle \in R$ ([illegible] R [illegible] R^{-1}-[illegible]).

[illegible] :[illegible] (1)

1. [illegible] R, S, [illegible] $R \cup S$ [illegible].
2. [illegible] R, S, [illegible] $R \cup S$ [illegible].
3. [illegible] R: [illegible] R [illegible]-[illegible], [illegible] $\langle x, y\rangle \in R$ [illegible] $x = y$.
4. [illegible] $\emptyset$ [illegible]? [illegible]-[illegible]? [illegible]? [illegible] A?

1. [illegible] - [illegible] $\langle x, y\rangle \in R \cup S$. [illegible] $\langle x, y\rangle \in R$ [illegible] $\langle y, x\rangle \in R \subseteq R \cup S$; [illegible] $\langle x, y\rangle \in S$.
2. [illegible] [illegible] - $R = \{\langle 1, 2\rangle\}$, $S = \{\langle 2, 1\rangle\}$ [illegible] ([illegible]), [illegible] $R \cup S$-[illegible] $\langle 1, 2\rangle, \langle 2, 1\rangle$ [illegible] $\langle 1, 1\rangle$.
3. [illegible] - [illegible] $\langle x, y\rangle \in R$. [illegible] $\langle y, x\rangle \in R$, [illegible]- [illegible] ($x\,R\,y$ [illegible] $y\,R\,x$) [illegible] $x = y$.
4. $\emptyset$ [illegible], [illegible]-[illegible] [illegible] [illegible] ([illegible] - [illegible]). [illegible] A [illegible] $A = \emptyset$: [illegible] $a \in A$ [illegible] $\langle a, a\rangle \notin \emptyset$.

(2) [illegible] :[illegible]

5. [illegible] R, S [illegible] $R \cap S$ [illegible].
6. [illegible] R: R [illegible] $R = R^{-1}$.
7. [illegible] R, [illegible] $R \cup R^{-1}$ [illegible].
8. [illegible] R: [illegible] R [illegible]-[illegible], [illegible] R [illegible].

5. - [illegible] $\langle x, y\rangle, \langle y, z\rangle \in R \cap S$. [illegible] R-[illegible], [illegible] $\langle x, z\rangle \in R$; [illegible] $\langle x, z\rangle \in S$. [illegible] $\langle x, z\rangle \in R \cap S$.
6. - ($\Rightarrow$) [illegible] $\langle x, y\rangle \in R$ [illegible] $\langle y, x\rangle \in R$, [illegible] $\langle x, y\rangle \in R^{-1}$; [illegible]. ($\Leftarrow$) [illegible] $\langle x, y\rangle \in R$ [illegible] $\langle y, x\rangle \in R^{-1} = R$.
7. - [illegible] $\langle x, y\rangle \in R \cup R^{-1}$. [illegible] $\langle x, y\rangle \in R$ [illegible] $\langle y, x\rangle \in R^{-1}$; [illegible] $\langle x, y\rangle \in R^{-1}$ [illegible] $\langle y, x\rangle \in R$. [illegible] $\langle y, x\rangle \in R \cup R^{-1}$.
8. - [illegible] $x \, R \, y$ [illegible]-$y \, R \, z$. [illegible] 3, $x = y$ ([illegible]-$y = z$). [illegible] $x \, R \, z$ [illegible] $y \, R \, z$, [illegible].

$$F = \{X \subseteq \mathbb{N} \mid 0 \in X\}.$$

1. $\bigcup F$
2. $\bigcap F$

1. $\bigcup F = \mathbb{N}$
2. $\bigcap F = \{0\}$

□□□□□□: □□□□□□ □□□□□□ □□ □□□□□□□

□□□□□□ □□□ $n \in \mathbb{N}$ □□ □□□□□□□:

$$A_n = \{k \in \mathbb{N} \mid k \geq n\}$$

□□□ □□□□□□□ $\mathcal{F} = \{A_n \mid n \in \mathbb{N}\}$.

□□□□□ □□:

1. $\bigcup \mathcal{F}$
2. $\bigcap \mathcal{F}$

□□□□□□:

1. □□□□□□: □□□□□□□ □□□□□ □□□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□. □□□□□ □□□ □-$A_0 = \mathbb{N}$. □□□ $\bigcup \mathcal{F} = \mathbb{N}$.
2. □□□□□□: □□□□□□□ □□□□□ □□□□□□□ □□□□□□□□□ □ □□□□□□□□□. □□□ $k \in \mathbb{N}$. □□□ $k \in \bigcap \mathcal{F}$? □□, □□ $k \notin A_{k+1}$ (□□□□□ $k < k+1$). □□□ □□□ □□ □□□□□ □□□□□ □□□□□□ □□□□□□. $\bigcap \mathcal{F} = \emptyset$.

[illegible]: [illegible]

[illegible] $n \in \mathbb{N}, n \geq 1$ [illegible]:

$$B_n = \{k \in \mathbb{N} \mid n \text{ [illegible] } k\}$$

([illegible] B_n [illegible] n). [illegible] $\mathcal{G} = \{B_n \mid n \geq 1\}$.

[illegible]:

1. $\bigcup \mathcal{G}$
2. $\bigcap \mathcal{G}$

[illegible]:

1. [illegible]: B_1 [illegible] 1-[illegible] ([illegible], [illegible] $k \geq 1$). [illegible] $B_1 \in \mathcal{G}$-[illegible], [illegible] B_1. [illegible] B_1-[illegible]. [illegible] $\bigcup \mathcal{G} = B_1 = \{1, 2, 3, \dots\}$.
2. [illegible]: [illegible] k [illegible] -[illegible] [illegible] n. [illegible] k [illegible], [illegible] $n = k + 1$ [illegible] k ([illegible] $k > 0$). [illegible] $\bigcap \mathcal{G} = \emptyset$.

[illegible]: [illegible]

[illegible] $A = \{1,2,3\}$. [illegible] $\mathcal{H}$ [illegible]-[illegible] A [illegible] 2 [illegible].

$$\mathcal{H} = \{X \in \mathcal{P}(A) \mid |X| = 2\}$$

[illegible] $\mathcal{H}$.

1. [illegible]:

$$\mathcal{H} = \{\,\{1,2\},\{1,3\},\{2,3\}\}$$

2. [illegible]:

$$\bigcup \mathcal{H} = \{1,2\} \cup \{1,3\} \cup \{2,3\} = \{1,2,3\} = A$$

3. [illegible]: [illegible]:
   - 1 [illegible] $\{\mathbf{2},\mathbf{3}\}$-[illegible].
   - 2 [illegible] $\{\mathbf{1},\mathbf{3}\}$-[illegible].
   - 3 [illegible] $\{\mathbf{1},\mathbf{2}\}$-[illegible].
   - [illegible] $\bigcap \mathcal{H} = \emptyset$.

[illegible]

[illegible]: [illegible]:

1. $\forall A,B,C(A\cup(B\cap C)=(A\cup B)\cap(A\cup C))$
2. $\forall A,B,C(A\cup(B\setminus C)=(A\cup B)\setminus(A\cap C))$
3. $\forall A,B,C(A\setminus(B\cup C)=(A\setminus B)\cap(A\setminus C))$

1. [illegible]

$$\begin{aligned} x\in A\cup(B\cap C) &\iff x\in A\ \vee\ (x\in B\wedge x\in C)\\ &\iff (x\in A\vee x\in B)\wedge(x\in A\vee x\in C)\\ &\iff x\in(A\cup B)\cap(A\cup C)\end{aligned}$$

2. [illegible]

[illegible]:

$$A=\{1,2\},B=\{2,3\},C=\{3,4\}$$

$$A\cup(B\setminus C)=\{1,2\}\cup\{2\}=\{1,2\}$$

$$(A\cup B)\setminus(A\cap C)=\{1,2,3\}\setminus\{2\}=\{1,3\}$$

3. [illegible]

$$\begin{aligned} x\in A\setminus(B\cup C) &\iff x\in A\wedge x\notin(B\cup C)\\ &\iff x\in A\wedge(x\notin B\wedge x\notin C)\\ &\iff (x\in A\wedge x\notin B)\wedge(x\in A\wedge x\notin C)\\ &\iff x\in(A\setminus B)\cap(A\setminus C)\end{aligned}$$

[illegible]:

[illegible] U.

1. [illegible]: [illegible] $A \subseteq U$ [illegible]

$$U \setminus (U \setminus A) = A.$$

2. [illegible]: [illegible] $A, B \subseteq U$ [illegible]

$$A \subseteq B \iff (U \setminus B) \subseteq (U \setminus A).$$

3. [illegible] U-[illegible]:

$$U \setminus (A \cup B) = (U \setminus A) \cap (U \setminus B),$$
$$U \setminus (A \cap B) = (U \setminus A) \cup (U \setminus B).$$

[illegible]: [illegible] B [illegible] - [illegible] $A = \emptyset$

[illegible]. [illegible], [illegible] $A \setminus B$ [illegible] B, [illegible] $A = \emptyset$.

1. [illegible] A, [illegible] B [illegible] $A \setminus B = \emptyset$, [illegible] $A = \emptyset$.
2. [illegible] A, [illegible] B [illegible] $A \setminus B \subseteq B$, [illegible] $A = \emptyset$.
3. [illegible] A, [illegible] B [illegible] $A \setminus B = A$, [illegible] $A = \emptyset$.
4. [illegible] A, [illegible] B [illegible] $A \setminus B \subseteq A \cap B$, [illegible] $A = \emptyset$.
5. [illegible] A, [illegible] B [illegible] $A \subseteq A \setminus B$, [illegible] $A = \emptyset$.

[illegible]: [illegible] [illegible] C [illegible] [illegible] - [illegible] $A = B$

[illegible] [illegible]. [illegible] [illegible], [illegible] [illegible] [illegible] $A \setminus C$ [illegible]-$B \setminus C$ [illegible] [illegible] [illegible] C, [illegible] $A = B$.

1. [illegible] A, B, [illegible] [illegible] C [illegible] $A \setminus C = B \setminus C$, [illegible] $A = B$.
2. [illegible] A, B, [illegible] [illegible] C [illegible] $A \setminus C \subseteq B \setminus C$ [illegible] $B \setminus C \subseteq A \setminus C$, [illegible] $A = B$.
3. [illegible] A, B, [illegible] [illegible] C [illegible] $A \setminus C \subseteq B$ [illegible] $B \setminus C \subseteq A$, [illegible] $A = B$.
4. [illegible] A, B, [illegible] [illegible] C [illegible] $A \setminus (B \cup C) = B \setminus (A \cup C)$, [illegible] $A = B$.
5. [illegible] A, B, [illegible] [illegible] C [illegible] $(A \setminus C) \setminus B = (B \setminus C) \setminus A$, [illegible] $A = B$.

[illegible]: [illegible] B [illegible] - [illegible] $A=\emptyset$

[illegible]. [illegible], [illegible] $A\Delta B$ [illegible] B, [illegible] $A=\emptyset$.

1. [illegible] A, [illegible] B [illegible] $A\Delta B = B$, [illegible] $A=\emptyset$.
2. [illegible] A, [illegible] B [illegible] $A\Delta B \subseteq B$, [illegible] $A=\emptyset$.
3. [illegible] A, [illegible] B [illegible] $A \subseteq A\Delta B$, [illegible] $A=\emptyset$.
4. [illegible] A, [illegible] B [illegible] $A\Delta B = A\cup B$, [illegible] $A=\emptyset$.
5. [illegible] A, [illegible] B [illegible] $(A\Delta B)\cap A = \emptyset$, [illegible] $A=\emptyset$.

[illegible]: [illegible] C [illegible] - [illegible] $A=B$

[illegible]. [illegible], [illegible] $A\Delta C$ [illegible] $B\Delta C$-[illegible] C, [illegible] $A=B$.

1. [illegible] A,B, [illegible] C [illegible] $A\Delta C = B\Delta C$, [illegible] $A=B$.
2. [illegible] A,B, [illegible] C [illegible] $A\Delta C \subseteq B\Delta C$ [illegible] $B\Delta C \subseteq A\Delta C$, [illegible] $A=B$.
3. [illegible] A,B, [illegible] C [illegible] $(A\Delta C)\Delta(B\Delta C)=\emptyset$, [illegible] $A=B$.
4. [illegible] A,B, [illegible] C [illegible] $(A\Delta C)\Delta(B\Delta C)\subseteq C$, [illegible] $A=B$.
5. [illegible] A,B, [illegible] C [illegible] $(A\Delta C)\cap C=(B\Delta C)\cap C$, [illegible] $A=B$.

1. [illegible] A, B, C, [illegible]:

$$C \cap (A \Delta B) = (C \cap A) \Delta (C \cap B).$$

2. [illegible] A, B, C, [illegible] ([illegible]):

$$(A \Delta B) \Delta (A \Delta C) = B \Delta C.$$

3. [illegible] A, B, [illegible]:

$$A \subseteq B \iff A \Delta B \subseteq B \setminus A.$$

4. [illegible] A, B, C, [illegible]:

$$(A \times C) \Delta (B \times C) = (A \Delta B) \times C.$$

5. [illegible] A, B, C, D, [illegible]:

$$(A \times C) \Delta (B \times D) \stackrel{?}{=} (A \Delta B) \times (C \Delta D).$$

[illegible]

1. [illegible]:

$$A \setminus \bigcup F = \bigcap \{A \setminus X \mid X \in F\}.$$

2. [illegible]:

$$A \subseteq \bigcap F \iff \forall X \in F \ (A \subseteq X).$$

3. [illegible]:

$$A \subseteq B \iff \forall x \ (x \in A \Rightarrow x \in B),$$

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C),$$

$$(A \Delta B) \setminus C = (A \setminus C) \Delta (B \setminus C).$$

4. [illegible]:

$$(\bigcup F \subseteq \bigcup G \ \wedge \ \bigcap G \subseteq \bigcap F).$$

1. : $\mathcal{F}$ A

$$\bigcup\{X \setminus A \mid X \in \mathcal{F}\} = \left(\bigcup \mathcal{F}\right) \setminus A$$

2. : $\mathcal{F}$ A

$$A \setminus \bigcup \mathcal{F} = \bigcap\{A \setminus X \mid X \in \mathcal{F}\}.$$

3. : $\mathcal{F}$ S :

   1. $x \in S \iff \left(\forall X \in \mathcal{F}\ (x \in X)\right)$
   2. T , $T \subseteq S$.